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We
Make
Possible

Scissor Lift

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NIHVA Technologies Private Limited

Established in 2012 in Pune (Maharashtra, India), we "NIHVA Technologies Pvt. Ltd." are the prominent manufacturer, importer, exporter of a comprehensive assortment of Hangar Door, Material Handling Equipments, Rolling Shutter, Sliding Roof and much more. The offered products are highly appreciated among our customers for their attributes like eye-catching design, waterproof, easy to maintain, long lasting, moisture resistance and easy to use. Further, our offered product is checked on well-set parameters of quality in order to provide a perfect range at the clients' end. we also service provider of Door Installation Service.

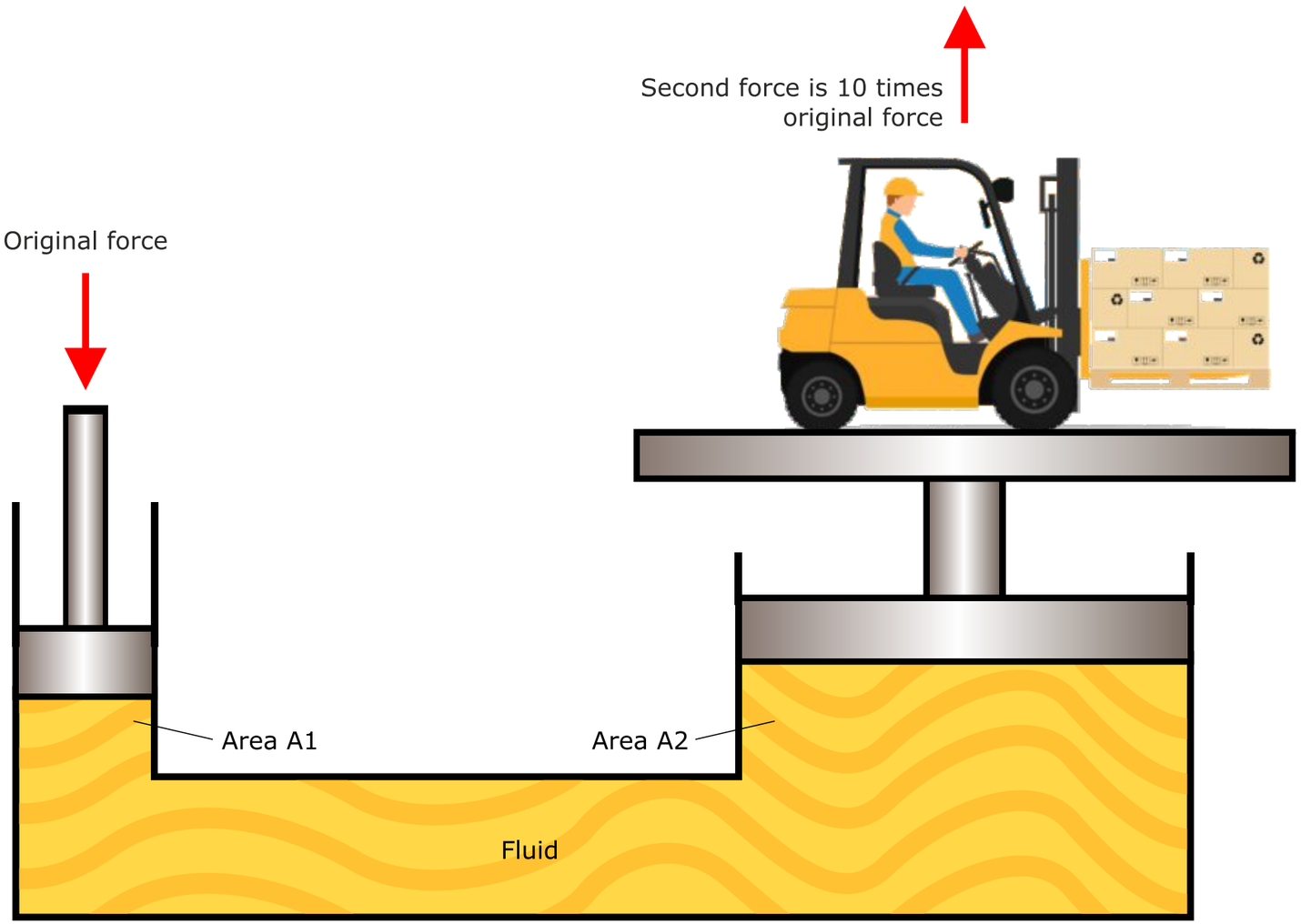
Introduction

Future Industries offers a wide range of Lift Tables in advanced features to cater to vast requirements of clients. The NIHVA Lift Table provided by us is safe & efficient and ways to endless Material Handling Applications at Distribution Centre, Logistics Companies, Industries etc. Our Lift Tables are popular in the market because we don’t compromise on the quality and always endeavor to meet International Standards by stringent quality checking and using State-of-the-art Technologies in the manufacturing process.



Working Principle of Hydraulic Scissor Lift

Pascal's principle



The concept of a scissors lift with hydraulic power comes from Pascal's law applied in car jacks and hydraulic rams which states that "pressure exerted anywhere in a conformed incompressible fluid is transmitted equally in all directions throughout the fluid such that the pressure ratio remains the same" (Hydraulics online 2013).

Why Scissor Lift ?

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The scissors lift is an elevator with a system of levers and hydraulic cylinders on which the metal platform is capable of moving in the vertical direction. This is achieved by using of linked, folding supports in a crisscross pattern, called scissor mechanism.

Scissors lift is an integral part of most of the workshops and building objects. The key advantage of lifts is that they even offer the best way to organise a technological and industrial process. Besides, almost all lifts give the possibility to change the place of their installation without much effort, which is important in the frequently changing conditions in the production process these days.

The scissors lift has a table surface where the weight can be placed. The rising of a platform is carried out due to work of a hydraulic cylinder. The management of the Lift is carried out by the control panel which has four buttons: the main switch, platform Up, Down, and emergency switch.

As rising speed uniform and slow (on average 30 mm per second).

Types of Scissor Lifts

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The design of scissor lifts can be divided into two main types:

1. Permanent (PIT MOUNTED OR FLOOR MOUNTED: The permanent scissor lifts are: scissor raise platforms, track lifting platform, loading and unloading platforms.



2. Portable (MOBILE): The portable scissor lifts divided into: several mobile lifting platforms, a couple of tractor-lift platforms, AC-DC dual-use working out with a platform, lightweight aluminium & Stainless-steel lift platform, working out with height from 1-10m. As the scissor lift divided into: electro-hydraulic & manual-hydraulic.



Scissor Lift

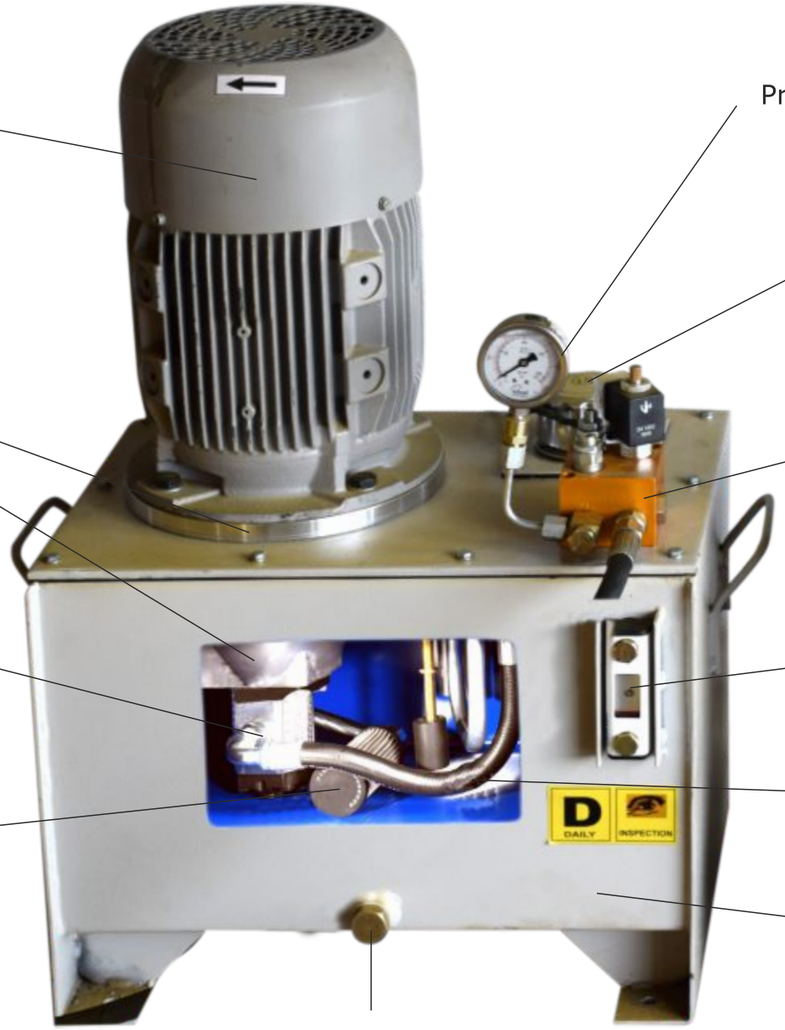
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Power Pack


AC Motor
Pressure Gaug
Oil filler breather
Bell Housing
Manifold Block
(Directional Solenoid Valve)
Gear Pump
Oil Level Indicator
Pump Pressure Line
Flexible Hose
Suction Strainer
Oil Reservoir
(Hydraulic grade-68)
Oil Drain Plug
D
DAILY
INSPECTION

Control Panel



Safety Features

Hydraulic Accessories



Pilot Operated Check Valves
Function of Pilot Operated Check Valves: Designed for use in applications requiring an actuator to be locked in position, Pilot Operated Check Valves allow free flow from the input port through the output port. Supplying a pilot pressure to the valve allows flow in the opposite direction.



Hose burst valve
The hose burst valve is a flow operated flat seat valve.
Therefore, it prevents uncontrolled movement in the consumer in the event of a hose burst.
The hose burst valve is open in the normal position and allows flow in both directions. If there is an excess of the pre-set actuating flow rate as a result of a hose burst, for example, the valve quickly closes and blocks the flow from port 1 to port 2.



Oil return line filter
Return Line Filter for cleaning the oil going through the filters.
The return line filters have reliably remove the contamination from the operating fluid. Following this filtration, the fluid flows back into the tank

Safety Features

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Electrical Accessories



Over load relay
A relay that opens a circuit when the load in the circuit exceeds a preset value, in order to provide over load protection; usually responds to excessive current, but may respond to excessive values of power, temperature, or other quantities. Also known as overload release.



M.C.B.
A miniature circuit breaker (MCB) automatically switches off electrical circuit during an abnormal condition of the network means in overload condition as well as faulty condition. Handling an MCB is electrically safer than a fuse.



Loto - Switch
A lockout device is a key or combination lock with a special tag with a worker’s name. The lock must be attached to an isolating device, circuit breaker, and/or switch to prevent turning on the energy source or equipment or releasing energy.

Safety Features

Mechanical Accessories

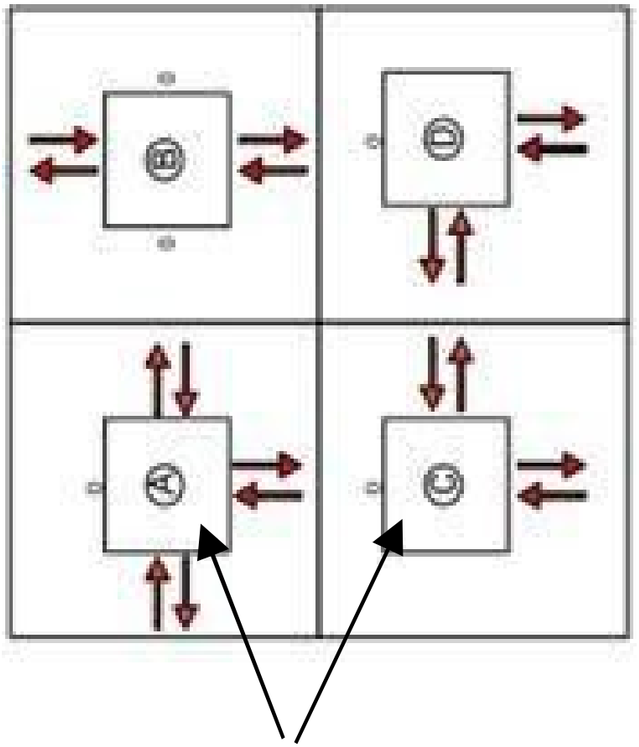


Safety fabric bellows

Bellows are used to cover criss-cross portion of scissor liftin operating condition to prohibit manual interference under the working platform.

Check List

SCISSOR LIFT TABLE - CHECK LIST			
Sr. No	**Description**	**Unit**	**Remark**
1	Type - Floor Mounted / Pit Mounted / Mobile		
2	Capacity - Kgs	Kg	
3	Platform Size - L X W (mm)	mm	
4	Lift height (Excluding close height)	mm	
5	Closed Height	mm	
6	Total height	mm	
5	Available Space (L X W)	mm	
6	No of lip required	mm	
7	Lip (Length X Width)		
8	Lip Operation (Manual / Hydraulic)		
9	Entry & exit side (please specify in detail)		
10	Load type (with pallet / without pallet)		
11	Type of platform (chequered plate / plain sheet) any other material please specify		
13	Color (NIHVA STD = GOLDEN YELLOW)		
14	Other requirement - please specify in details		
15	Operation through (Manual / AC OPERATED) Please specify the MHE details		



Entry / Exit Option

Inspection Report

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INSPECTION REPORT	
Product Name:	**SCISSOR LIFT**
Serial No.	
Specification:	
Customer:	
Customer Address:	
Challan No.:	
Inspected by:	
Dispatch Date:	

Sr No.	Part Description	Specification/ Dimension	Observation	Remark
1	Capacity			
2	Platform size			
3	Lip Size			
4	Shut height			
5	Maximum height			
6	Lift height			
7	Railing height			
8	Power pack			
9	With load pressure			
10	Without load pressure			
11	Total lifting time			
12	Oil leakages			
13	Electracal connection			
14	Paint			

Format No. NIHVA-PRD-F-07

Inspected by

ELECTRO HYDRAULICSCISSOR LIFT DO’s & DONT’s: -

DO’s: -

- Check all electrical connection.
- Check hydraulic connection.
- Ensure no oil leakage from joints.
- Ensure oil level on level indicator before use the lift.
- Before loading & unloading ensure lip rest in truck.
- Ensure any obstacle within motion part including up & down motion.
- While maintenance work under the lift ensure Emergency switch is pressed & maintenance stopper is in its position.
- Under the brake down condition ensure LOTO switch with its proper tag.

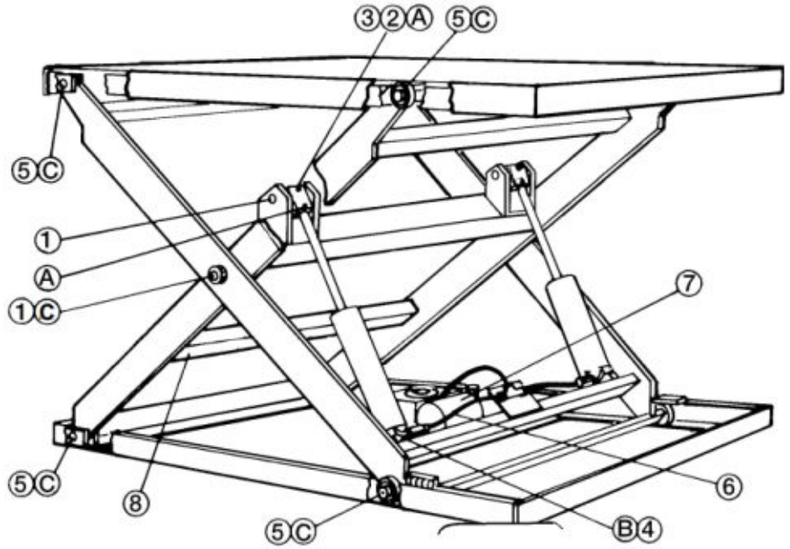
DONT’s: -

- Lift should not be overloaded beyond the rated load capacity.
- Don’t loose the oil hoses during lift is in its max position.
- Do not operate damaged lift.
- Don’t keep the feet at edges of lift platform while lift is lowering.
- Don’t change lowering valve setting without supervision or any assistance.
- In case of emergency power cut off & on do not lower the lift directly slightly lift it upward & then lower the lift.
- Do not disengaged chain provided when lip is rested in truck.
- Don’t apply any manual stopper in roller track.

OPERATING INSTRUCTIONS

a) Check all electrical connection for nobody short or phase short Check hydraulic connections are tightened properly to ensure no oil leakage from joints.
b) Connect (3 phase 415V) supply to the panel through cable top.
c) Switch ON the RED LOTO SWITCH on panel & ensure Emergency push button is in unlock position.
d) Use Platform push button for platform lift & lowering.
e) Use Lip push button for lift & lowering.
f) In case emergency, press emergency push button provided on push button station.
g) Before lifting Scissor, check motor rotates in clock wise direction.
h) For controlling lowering speed, adjust valve provided on manifold.
i) In case of failure & scissor lift platform is at top position, loosen the threads of manual override provided in the solenoid valve after reaching the platform at down position again retighten the same.

LUBRICATION & ROUTINE MAINTENANCE INSTRUCTIONS



➢ Lubrication Points

Lubricate points as mentioned below,

A. Points noted with ‘A’, apply grease lubrication.

B. Points noted with ‘B’, apply oil lubrication.

C. Points noted with ‘C’, no lubrication required (Pre-Lubricated &sealed)

Ensure all Moving parts like Wheel, Flap Joint, Rollers etc. are greased.

➢ Inspection Check Points

a) Visually inspect all jack pins, pivot pins to ensure that the set screw has not loosened, allowing pins to move.

b) Confirm Hex Nut Bolt, Allen Bolt -Pivot Pin Locking are Fixed firmly.

c) Confirm Grub screw, Cap Washer on Scissor Arm Joint Bolt are fixed & are tightened firmly.

d) Confirm Tightening of Rollers & its free movement.

e) Confirm all Bolts at Scissor Arm are tightened.

f) Visually inspect all lift arm rolling bearings to ensure that bearing bolts are tight.

g) Visually inspect all hydraulic hoses and fittings to ensure there are no cut on hoses or leaking fittings.

h) Visually inspect all electrical lines to ensure there are no cut wires or loose cords.

i) Visually inspect the entire lift structure.

j) These recommendations are applicable to the equipment considering daily usage of approx. 15-to 20 times.

If usage is more frequent, the same time intervals for maintenance should be adjusted accordingly.

➢ WEEKLY

a) Do greasing / oiling of all moving joints.

➢ MONTHLY

a) Check hydraulic pipeline for leak proof working.

b) Check oil level in hydraulic tank.

c) Check cylinder seals & ‘O` rings for leak proof working.

➢ YEARLY

a) Change hydraulic oil.

b) Check main shaft seal & shaft ‘O` ring for leak proof working.

Hydraulic oil to be used – Enclo – 68 OR Servo – 68.

GUIDE LINES IN CASE OF OIL LEAKAGE

a) Leakage at Piping –

- Tighten Joints.
- Replace Dowty Seal at Mounting

b) Leakage in Cylinder-

- Replace Seals.

c) Replace Oil after Every 3 Years-

- Drain the Oil from the System.
- Clean filter, Tank elements, and strainers before refilling.

TROUBLE SHOOTING GUIDELINES

i. If lift will not raise:

a. Check electric circuit on new installations of 3-phase motors, if lifts do not start rising in approximately 10 seconds, reverse two power supply leads to reverse pump rotation. (if motor direction is reverse)
b. Check oil level. With lift in full up position, oil level should be Approximately 1 1⁄4" above tank bottom.

ii. Lift raises slower than specified rate or will raise only partial load:

Note: Allowance must be maintained for slight speed variations (due to temperature which changes the viscosity of the hydraulic oil varies)
a. Check line voltage under load condition. Low voltage affects capacity of equipment.
b. If foam is visible in tank oil, check for a loose connection in suction line between pump and tank. Tighten connections.

iii. Lift slowly settles with lowering valve not energized:

a. Flush lowering valve by cycling lift up and down. This should be done with no load.
b. If lift continues to settle, replace down valve cartridge.
c. If lift continues to settle, replace pump.

iv. If Lift will not lower:

a. Look for mechanical restrictions.
b. Test valve coil for operation. Check voltage at coil.
c. If voltage is ok, replace coil.

v. If lift will not raise to full vertical travel:

a. Check for low oil level if necessary, add oil.

vi. If lift does not lower smoothly:

a. Run lift up and down under load condition for some time to purge air.

GENERAL SAFETY INSTRUCTIONS

Use trained personnel for loading & unloading of equipment. The equipment must be properly handled when transferring from the unloading area to final site to prevent damage.

DO NOT operate equipment unless it is properly anchored. It may cause serious injury or death.

a) Install all safety devices, guards and guarding prior to equipment start-up.
b) Only Trained & Authorized person must use the lift.
c) Always use mechanical stopper if you are going to work in elevated.
d) Keep working area free from oil to Avoid Slipping/Fire hazards.
e) Do Not Exceed the Lift Capacity than rated.
f) Before cleaning or servicing disconnect power supply
g) As the lift moves up and down, “pinch points” are formed. Keep hands, feet, hair, loose clothing, etc. Away from these pinch points. If any of these items become caught, may cause serious personal injury or death.
h) Ensure that the hydraulic, pneumatic and electrical lines will not be pinched by the lift as it rises or lowers. If a line becomes pinched, the lift may not work properly. A hose may break, the lift table may drop suddenly, and someone may injure. If an electrical line is broken, may cause personal injury or death due to electrical shock.
i) Read the motor nameplate and record the motor full load current. Consult local electrical codes for power supply fusing requirements. To avoid fire hazards, follow these fusing requirements.



Hydraulic Cylinder Test Procedure

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The conventional way of testing the integrity of the piston seal in a double-acting cylinder is to pressurize the cylinder at the end of stroke and measure any leakage past the seal. This is commonly referred to as the "end-of-stroke bypass test". The major limitation of the end-of-stroke bypass test, is that it generally doesn't reveal ballooning of the cylinder tube caused by hoop stress as a result of under-designed cylinder wall thickness or reduction of wall thickness through excessive honing. The ideal way to test for ballooning of the cylinder tube is to conduct a piston-seal bypass test, mid-stroke. The major difficulty with doing this is that the force developed by the cylinder has to be mechanically resisted, which in the case of large diameter, high-pressure cylinders is impractical. However, a mid-stroke bypass test can be conducted hydrostatically using the intensification effect. The necessary circuit is shown in Figure 1 below.

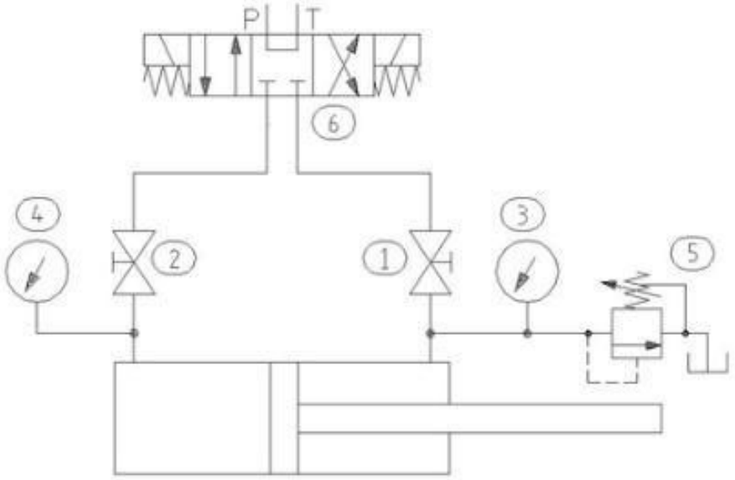


Figure 1. Hydraulic cylinder test circuit.

Test procedure

The procedure for conducting the test is as follows:

1. Secure the cylinder with its service ports up.
2. Fill both sides of the cylinder with clean hydraulic fluid through its service ports.
3. Connect ball valves (1) and (2), gauges (3) and (4), relief valve (5) and directional control valve (6) as shown in Figure 1.
4. With ball valves (1) and (2) open, stroke the cylinder using the directional control valve (6) multiple times to remove all remaining air from both sides of the cylinder.
5. Position the piston rod mid-stroke and close ball valve (2).
6. With the adjustment on the relief valve (5) backed out, direct flow to the rod side of the cylinder.
7. Increase the setting of relief valve (5) until the cylinder's rated pressure is seen on gauge (3).
8. Close ball valve (1) and centre directional control valve (6).
9. Note: it is assumed that the hydraulic power unit used to conduct the test has its own over-pressure protection - not shown in Figure 1.
10. Record the respective pressure readings on gauges (3) and (4) and monitor any change over time.

If the ratio of effective area between the piston and rod side of the cylinder is 2:1, then if the rod side of the cylinder has been pressurized to 3,000 PSI, gauge (2) on the piston side should read 1,500 PSI. If the differential pressure across the piston is not maintained, this indicates a problem with the piston seal or tube.

Thank You !